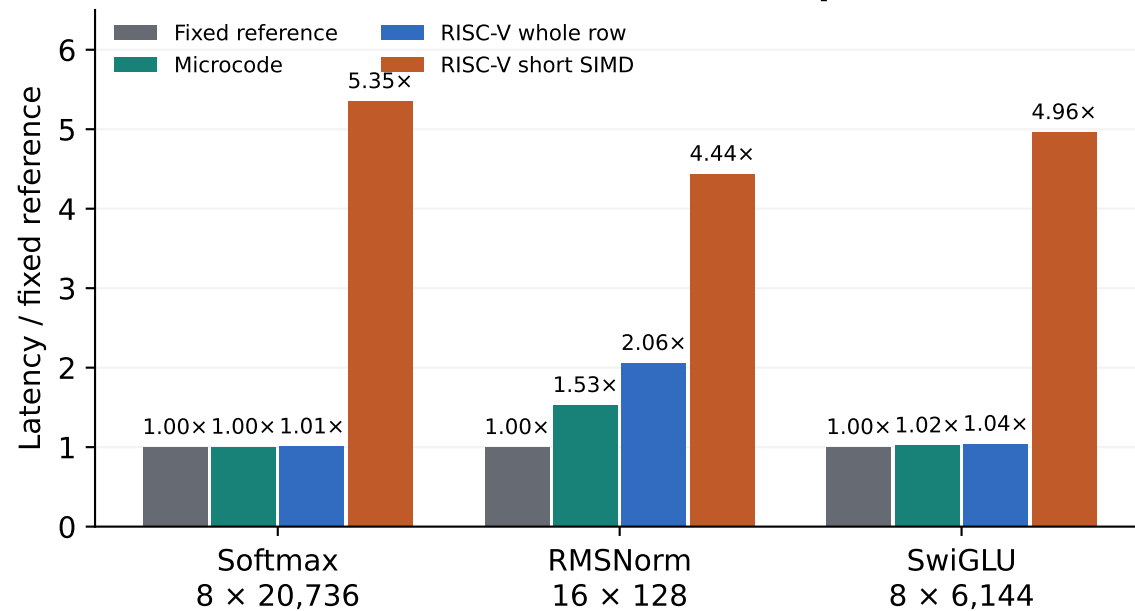


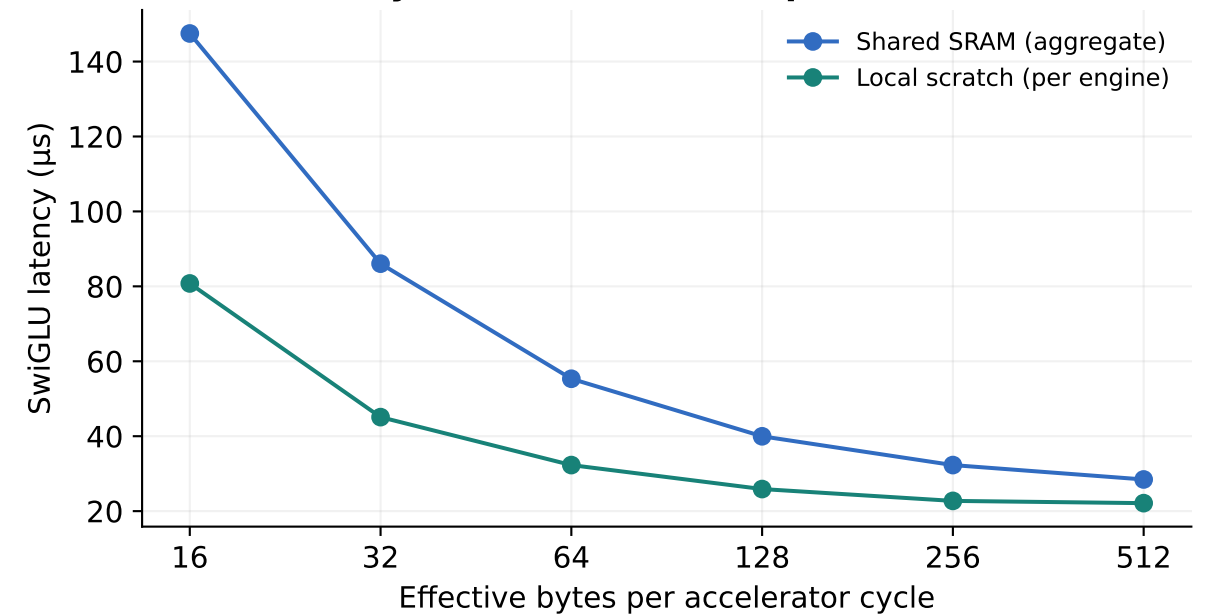
Programmability depends on command granularity

Analytical estimates • Qwen3.5-2B kernel shapes • same arithmetic and memory for each controller

Short commands can starve the datapath

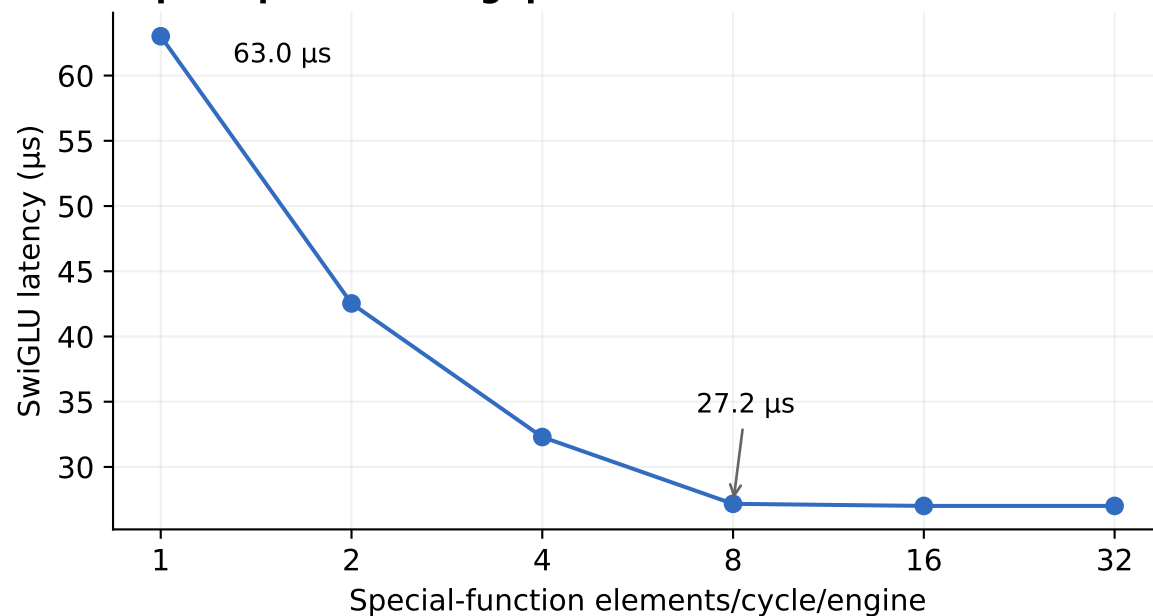


More memory bandwidth can help



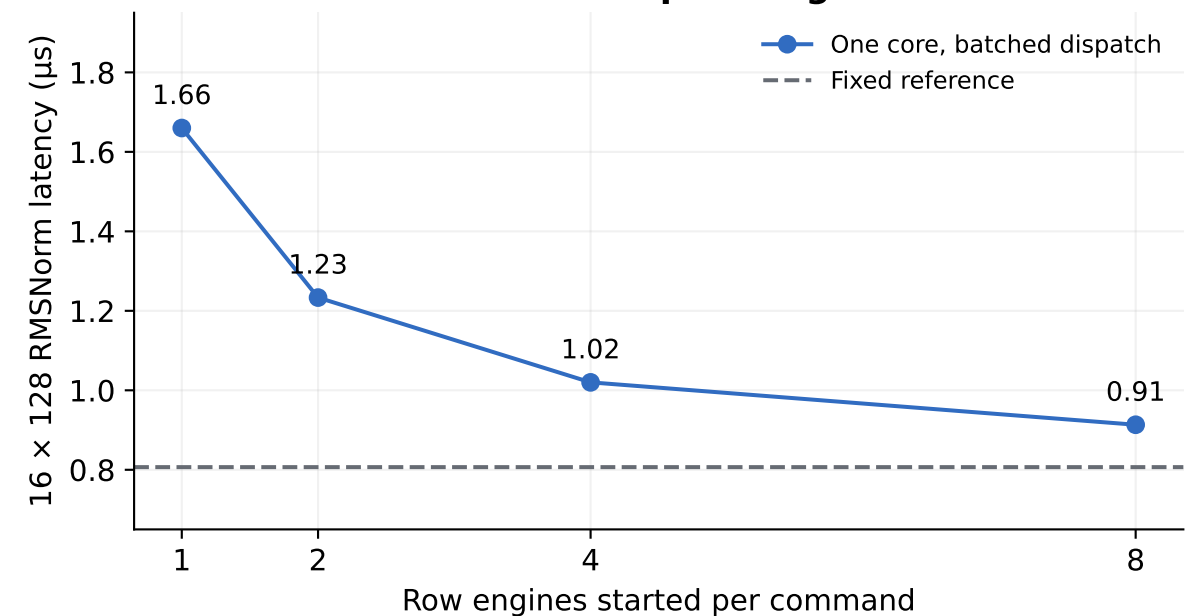
Each curve varies one bandwidth; all other settings stay fixed.

Exp/reciprocal throughput beats wider SIMD here



16 → 128 lanes alone leaves this workload at ≈32.3 μs.

Batch small rows before replicating cores

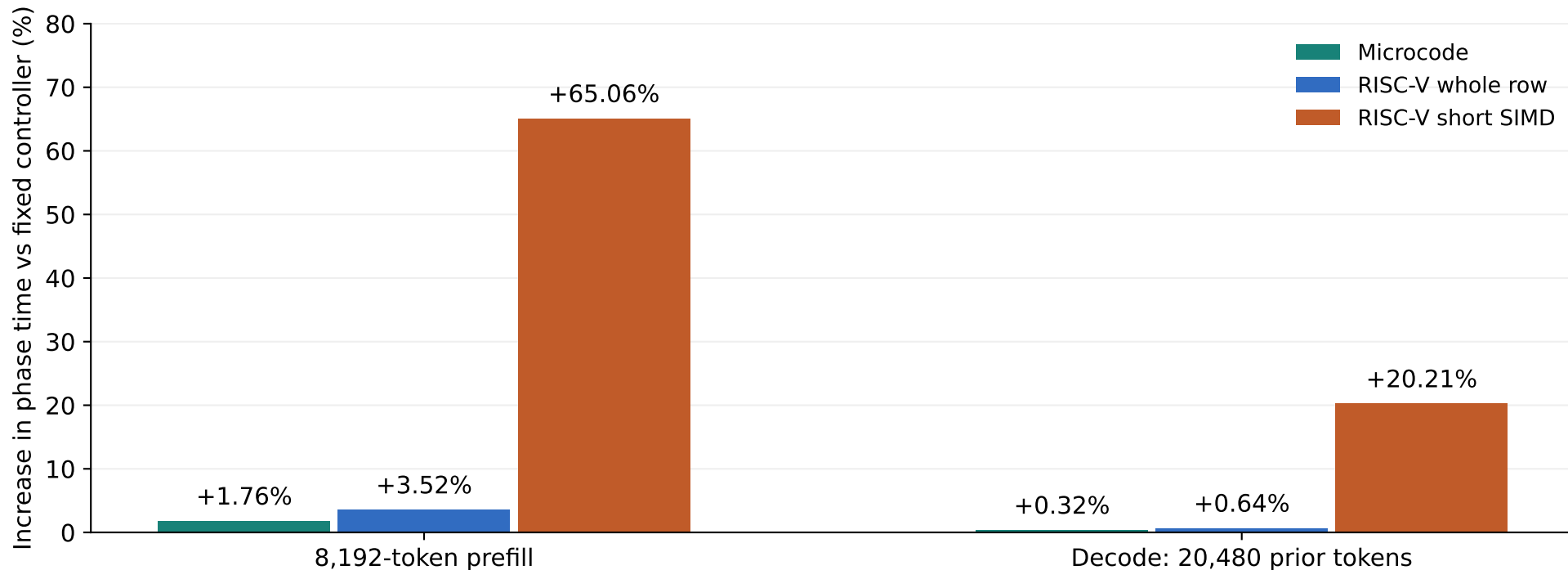


Batching requires address generators and completion tracking.

Reference: 8 engines × 32 lanes, 300 MHz, 64 KiB scratch/engine, 4-cycle RISC-V issue.
No RTL, area, power, instruction traces, or numerical validation. Resource sweeps do not hold area constant.

Most whole-model time is outside controller dispatch

Exploratory projection into the captured Qwen3.5-2B graph



Identical compute and memory within each controller comparison. Unsupported nodes retain inherited timings.
FP32 vector temporaries; inherited W4/A8 matrix model. Conversion, bank conflicts, and new HBM allocation omitted.
Model-to-model sensitivity, not measured inference performance.